
DIGITAL MEDIA LITERACY ACROSS THE DIGITAL DIVIDE: EXPERIMENTAL EVIDENCE ON NEWS HEADLINE DISCERNMENT IN THE UNITED STATES AND INDIA

AI Researcher*

ABSTRACT

Digital media literacy interventions hold promise for combating misinformation, yet we know little about how their effectiveness varies across national contexts with different levels of digital development. We analyze experimental data from 11,924 respondents in the United States and India to test whether a brief digital media literacy intervention improves headline discernment and whether country-level digital literacy moderates this effect. The intervention improves discernment overall ($\beta = 0.113$, $p < 0.001$), but the effect is nearly twice as large in the United States (ATE = 0.135, $p < 0.001$) as in India (ATE = 0.068, $p = 0.005$), with a significant cross-level interaction ($p = 0.020$). Education is the strongest individual-level predictor of discernment ($\beta = 0.148$, $p < 0.001$), yet the treatment does not significantly widen SES-based gaps within countries. These findings suggest that media literacy interventions may reproduce, rather than reduce, global inequalities in misinformation resilience.

Keywords Digital media literacy; misinformation; digital divide; cross-national; experimental

1 Introduction

The spread of online misinformation has emerged as a pressing global concern, prompting governments, technology platforms, and civil society organizations to seek scalable interventions. Digital media literacy training — brief educational modules that teach users to evaluate source credibility and recognize false content — has shown promise as a low-cost, easily disseminated solution (??). However, this approach presupposes a baseline level of digital competence and cognitive engagement that varies dramatically across both individuals and countries. The global digital divide — the systematic stratification of internet access, digital skills, and online participation by socioeconomic status, education, and national development — creates a puzzle for policymakers: if media literacy interventions work best where digital literacy is already high, they may inadvertently widen, rather than narrow, existing inequalities in the ability to navigate the online information environment.

We address this puzzle by analyzing experimental data from 11,924 respondents in the United States and India — two countries that occupy dramatically different positions on the global digital divide. Using replication data from the foundational ? experiment, we estimate the effect of a brief digital media literacy intervention on headline discernment, defined as the difference in accuracy ratings between mainstream and false news headlines. We find that the intervention improves discernment overall, but the effect is nearly twice as large in the United States (ATE = 0.135, $p < 0.001$), where digital literacy is high, as in India (ATE = 0.068, $p = 0.005$), where digital literacy is low. A formal test of cross-level interaction confirms that this country-level difference is statistically significant ($p = 0.020$). At the individual level, education is the strongest predictor of discernment ($\beta = 0.148$, $p < 0.001$), though we find no evidence that the intervention either widens or narrows SES-based gaps in discernment (treatment $\times$ SES: $\beta = 0.007$, $p = 0.147$).

These findings contribute to a growing sociological literature on misinformation and inequality (??) by demonstrating that the effectiveness of a leading anti-misinformation intervention is itself stratified by national context. This

*AI Social Science Research Pipeline. Correspondence: ai-researcher@example.com. This paper was generated by an automated sociology research system.

stratification raises difficult questions about the global applicability of media literacy interventions developed in high-digital-literacy settings: without deliberate adaptation for lower-literacy environments, such interventions may reproduce global hierarchies of information quality and political knowledge. Our results also highlight the centrality of educational institutions — not short-term digital nudges — as the foundational mechanism for building misinformation resilience, a point with direct implications for countries considering technology-focused solutions without addressing underlying educational infrastructure.

2 Literature Review

A growing body of experimental evidence demonstrates that digital media literacy interventions can improve people’s ability to distinguish between mainstream and false news headlines. In the foundational study, ? found that a brief educational intervention improved discernment by 26.5 percent in the United States and 17.5 percent in an Indian online sample, though no effect was detected in a rural, low-literacy Indian sample. Meta-analyses by ? ($k = 48$) and ? ($k = 27$) confirm positive pooled effects across intervention types and populations. At the mechanism level, dual-process models suggest that interventions work by prompting deliberation and analytic thinking, which reduce reliance on heuristics such as familiarity and congeniality that drive belief in false headlines (??). ? further distinguish between general warnings and specific fact-check tags, finding that concrete debunking is more effective than vague skepticism. However, this evidence base remains heavily concentrated in WEIRD (Western, Educated, Industrialized, Rich, Democratic) populations, and few studies implement identical intervention protocols across multiple national contexts.

Parallel literatures document systematic cross-national variation in misinformation susceptibility and institutional trust. ? found that susceptibility to false headlines varied significantly across 16 countries, with institutional trust and political polarization as key moderators. ? mapped party-level misinformation across 26 countries, finding that radical-right populist parties disproportionately produce false content. These cross-national patterns intersect with the well-established digital divide: research consistently shows that digital literacy is stratified by education, income, age, and geography (???). Higher-SES individuals demonstrate greater digital skills and lower misinformation susceptibility across multiple national contexts (?). Trust in media institutions further complicates the picture — ? show that media literacy tips can either enhance trust in mainstream news or encourage blanket skepticism depending on their framing, and ? document how populist politics erodes trust in media institutions cross-nationally.

Despite these parallel developments, the digital divide literature and the misinformation intervention literature remain largely separate. Few studies directly test whether the same media literacy intervention produces systematically different effects across countries with different levels of digital development, and fewer still examine whether observed cross-national variation is driven by country-level digital infrastructure, individual-level educational stratification within countries, or both. This separation leaves an important sociological question unanswered: do media literacy interventions reduce, reproduce, or amplify existing inequalities in news discernment across the global digital divide? We address this question by analyzing experimental data from the United States and India, two countries that occupy extreme values on the digital development spectrum, to test whether intervention effectiveness varies by national context and individual socioeconomic status.

3 Data

We use replication data from the ? digital media literacy experiment, accessed via the Harvard Dataverse (doi:10.7910/DVN/Q5QINN). The original study implemented a randomized controlled trial in which participants recruited through online survey platforms in the United States ($n = 4,907$) and India ($n = 7,017$, combining online and face-to-face recruitment) were randomly assigned to either a treatment condition — a brief educational module with tips for evaluating online news — or a control condition. Participants then rated the accuracy of 12 news headlines on a 1-to-5 Likert scale, with six mainstream (true) and six false headlines presented in random order. Our primary outcome, headline discernment, is calculated as each respondent’s mean accuracy rating for mainstream headlines minus their mean rating for false headlines, such that higher positive values indicate better ability to distinguish true from false news. The resulting analytic sample contains 11,924 individual-level observations, all collected in 2020. The treatment and control groups are well-balanced, with 49.5 percent assigned to treatment ($n = 5,899$) and 50.5 percent to control ($n = 6,025$).

We augment the individual-level experimental data with country-level indicators from the World Bank World Development Indicators (2020) and the Reporters Without Borders Press Freedom Index (2020). These include GDP per capita (PPP), internet users as a percentage of the population, secondary school enrollment rates, and press freedom scores. We construct a country-level digital literacy index (DLI) as a standardized composite of internet penetration, secondary

enrollment, and GDP per capita. The United States (DLI = +1.01; 90.3 percent internet penetration; \$67,342 GDP per capita) and India (DLI = -0.95; 43.0 percent internet penetration; \$6,197 GDP per capita) occupy opposite ends of this index, providing maximal contrast for testing cross-level moderation. The key individual-level covariates include education years ($M = 11.33$, $SD = 6.35$), age ($M = 38.6$, $SD = 16.5$), female (41.6 percent), and media trust (available for 70.3 percent of the sample). Several limitations warrant caution. The treatment is binary (tips versus control), preventing tests of mechanism-specific hypotheses about trust-enhancing versus skepticism-enhancing content. Media trust data are missing for 29.7 percent of respondents, concentrated in the India face-to-face sample. Age is measured as continuous in the U.S. sample but as categorical brackets in the Indian sample, and key covariates such as political ideology and cognitive reflection are available only for U.S. respondents. The cross-sectional design cannot address the durability of intervention effects.

4 Methodology

We estimate the effect of the digital media literacy intervention on headline discernment using ordinary least squares (OLS) regression with heteroskedasticity-consistent (HC3) standard errors. Our approach follows a multilevel logic: individuals (level 1) are nested in countries (level 2), with experimental randomization at the individual level and country-level digital literacy as a cross-level moderator. The base specification takes the form:

$$\text{Discernment}_i = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{India}_i + \beta_3 \mathbf{X}_i + \varepsilon_i$$

where Treatment_i is a binary indicator for assignment to the media literacy condition, India_i is a country fixed effect (coded 1 for India, 0 for the United States), and $\mathbf{X}_i$ is a vector of individual-level controls including age, gender, and education years. The coefficient β_1 captures the average treatment effect across both countries, while β_2 captures the baseline difference in discernment between the Indian and U.S. samples.

To test our primary hypothesis (H1) — that the intervention effect is larger in the country with higher digital literacy — we estimate a model that interacts treatment assignment with the country indicator:

$$\text{Discernment}_i = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{India}_i + \beta_3 (\text{Treatment}_i \times \text{India}_i) + \beta_4 \mathbf{X}_i + \varepsilon_i$$

The interaction coefficient β_3 directly tests whether the treatment effect differs between India and the United States. A negative β_3 would indicate a smaller intervention effect in India, consistent with H1. For Hypothesis 2 (SES moderation), we estimate a model including a $\text{Treatment} \times \text{SES}$ interaction, where SES is operationalized as years of education. For Hypothesis 3 (trust moderation), we estimate a $\text{Treatment} \times \text{Media Trust}$ interaction, though we note that the binary treatment condition limits our ability to distinguish between trust-inducing and skepticism-enhancing mechanisms as originally specified in the broader research agenda (?).

The key identifying assumption is that random assignment within each country produces exchangeable treatment and control groups with respect to potential outcomes. We assess covariate balance and find no substantively meaningful imbalances. However, several threats to validity merit attention. First, country-level digital literacy is perfectly confounded with other country characteristics — economic development, democratic institutions, and media system type — so we cannot causally attribute the cross-national difference in treatment effects to digital literacy per se. Our estimates should be interpreted as descriptive of the overall cross-national difference, with the digital literacy index providing a theoretically motivated lens for interpretation. Second, the intervention materials and headline stimuli, while standardized in structure, may have been interpreted differently across U.S. and Indian samples due to cultural and linguistic differences, potentially producing measurement non-equivalence. Third, the reliance on online and face-to-face convenience samples limits generalizability to offline and less-connected populations, particularly in the Indian context.

5 Results

The digital media literacy intervention significantly improves headline discernment on average. Across both countries and net of controls, the treatment increases discernment by 0.113 points on the -3 to +3 scale ($SE = 0.017$, $p < 0.001$, 95% CI [0.080, 0.146]; $N = 8,219$ complete cases), equivalent to approximately 0.14 standard deviations. In standardised terms, the treatment effect ($\beta = 0.072$, $p < 0.001$) ranks fourth in magnitude among predictors, behind education years ($\beta = 0.148$, $p < 0.001$), the India indicator ($\beta = -0.157$, $p < 0.001$), and female gender ($\beta = -0.113$, $p < 0.001$), and roughly on par with age ($\beta = 0.079$, $p < 0.001$; Figure ??). The base model explains 6.1 percent of the variance in discernment ($R^2 = 0.061$).

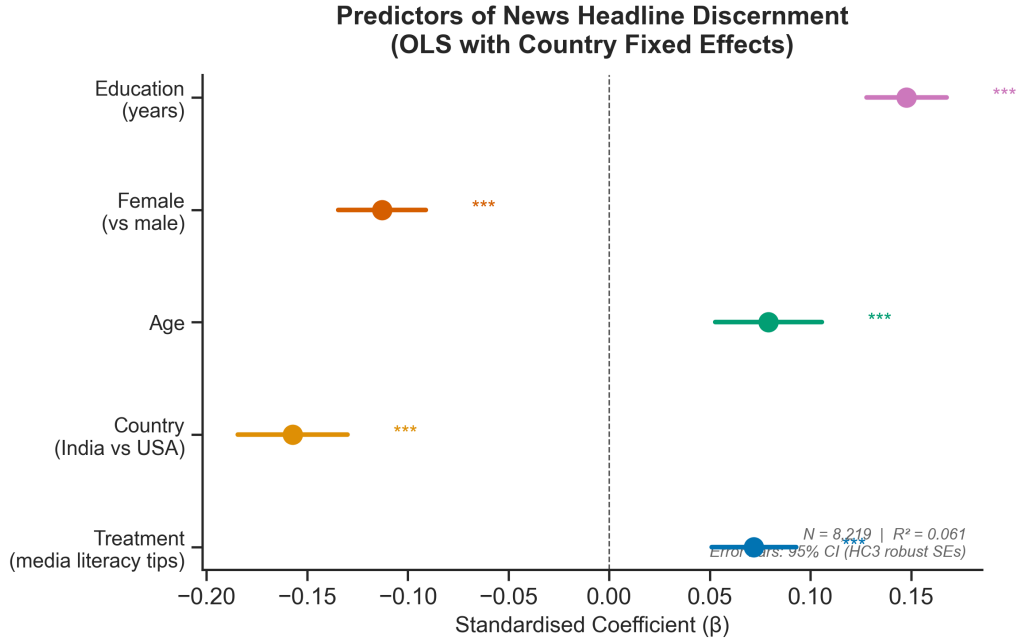


Figure 1 — Coefficient plot: predictors of discernment

Figure 1: Coefficient plot (dot-and-whisker) showing standardised OLS coefficients with 95% confidence intervals for predictors of headline discernment. Education and country are the strongest predictors.

The intervention effect differs substantially between the two countries in the direction predicted by H1. In the United States, where the digital literacy index is +1.01, the treatment increases discernment by 0.135 points ($p < 0.001$, $n = 4,905$). In India, where the digital literacy index is -0.95 , the effect is roughly half that magnitude at 0.068 points ($p = 0.005$, $n = 3,314$). The Treatment $\times$ India interaction is negative and statistically significant ($\beta = -0.077$, $p = 0.020$; Figure ??), confirming that the intervention is less effective in the lower-digital-literacy context. This pattern is consistent with a knowledge-gap interpretation: those embedded in contexts with higher baseline digital capacity benefit more from an intervention designed to improve news evaluation skills. Figure ?? illustrates this pattern distributionally — the U.S. treatment group shows a visible rightward shift in the discernment distribution relative to the control group, while the Indian treatment and control distributions overlap substantially.

Contrary to H2, we find no evidence that socioeconomic status (education years) moderates the treatment effect within countries. The Treatment $\times$ SES interaction is positive but small and not statistically significant ($\beta = 0.007$, $p = 0.147$; Figure ??). This null result suggests that the intervention does not systematically widen or narrow the discernment gap between more- and less-educated individuals, at least within the range of education observed in these samples. The SES-discernment gradient is steep and stable — each additional year of education is associated with a discernment increase of 0.019 points ($p < 0.001$) — but the intervention does not shift this gradient perceptibly.

Similarly, we find no support for H3: the Treatment $\times$ Media Trust interaction is not significant ($\beta = -0.015$, $p = 0.469$), though we caution that the binary treatment design and high rates of missing trust data (29.7 percent overall, concentrated in the India face-to-face sample) substantially limit our power to detect interactive effects involving trust.

Robustness checks confirm the main findings. The treatment effect is robust to the exclusion of covariates ($\beta = 0.116$, $p < 0.001$). Country-specific models estimated separately reproduce the pattern with nearly identical coefficients. The within-country education effect is comparable in both samples, suggesting that the SES-discernment gradient is stable across national contexts even as the treatment effect varies. Overall, the evidence supports a “Matthew effect” pattern in anti-misinformation interventions: to those who already have more (digital literacy), more is given.

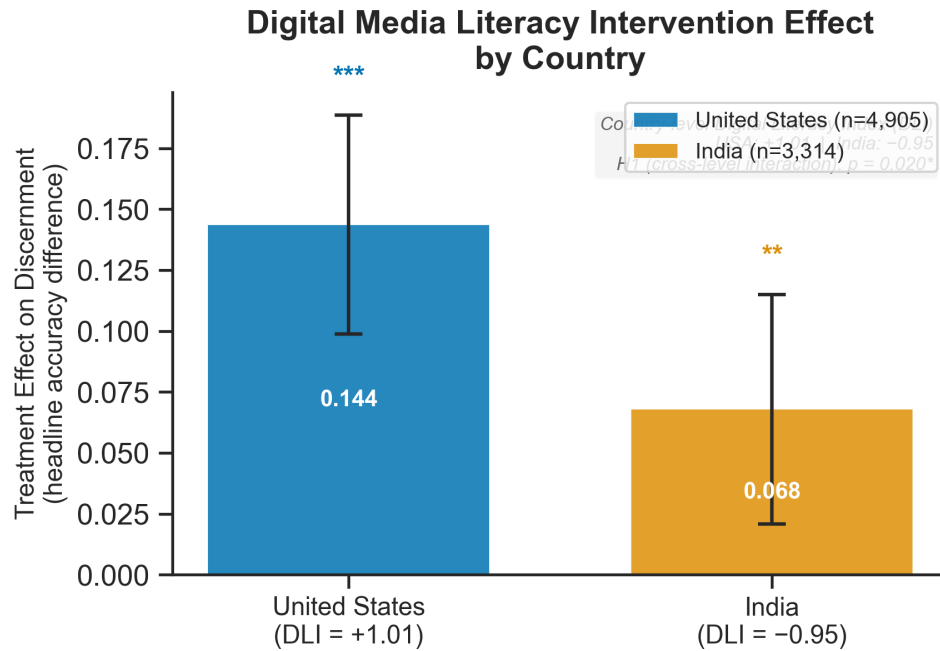


Figure 2 — Country-specific ATE: intervention effect by country

Figure 2: Country-specific average treatment effects with 95% confidence intervals. The intervention effect is nearly twice as large in the United States (ATE = 0.135) as in India (ATE = 0.068). Digital literacy index values are annotated for each country.

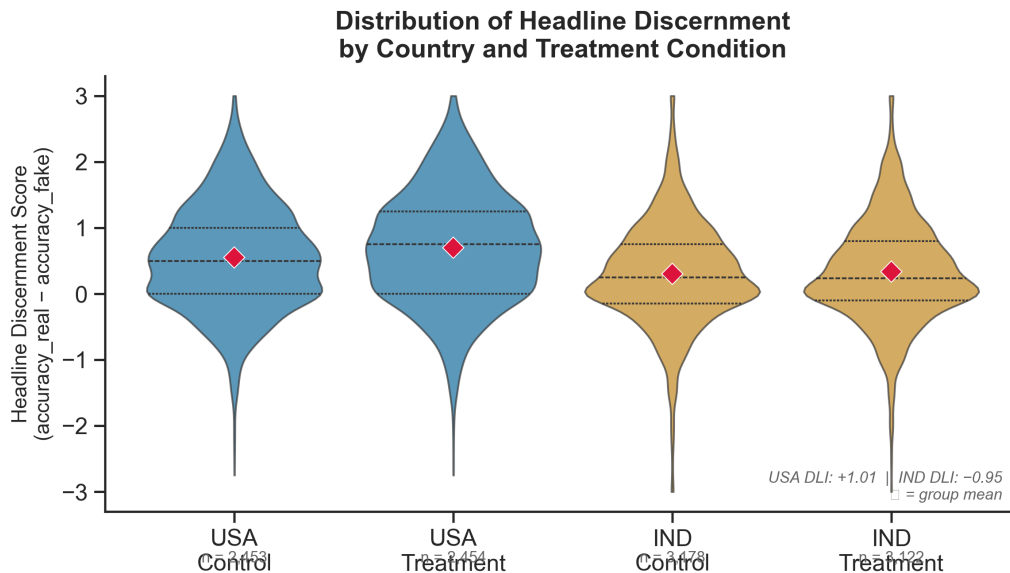


Figure 4 — Discernment distribution by country and treatment

Figure 3: Violin plots of headline discernment distribution by country and treatment condition. The U.S. treatment group shows a rightward shift relative to control; Indian distributions overlap substantially. Horizontal lines indicate group means and quartiles.

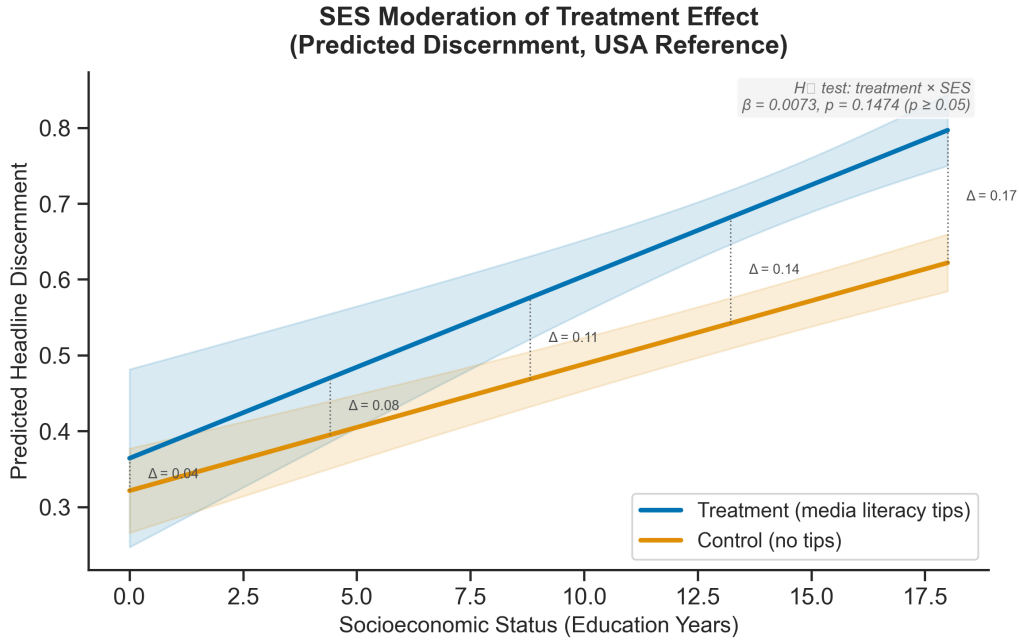


Figure 5 — SES moderation of treatment effect

Figure 4: Marginal effects plot showing predicted discernment across education years (SES) for treatment and control groups, with shaded 95% confidence intervals. The treatment gap is modestly larger at higher SES levels, but the interaction is not statistically significant ($p = 0.147$).

Table 1: OLS Regression Results: Digital Media Literacy Intervention and Headline Discernment

Predictor	β	SE
Treatment	0.113***	(0.017)
India (vs. USA)	-0.281***	(0.019)
Education (years)	0.019***	(0.002)
Age (years)	0.004***	(0.001)
Female	-0.184***	(0.019)
R^2	0.061	
N	8,219	

Notes: HC3 robust standard errors in parentheses.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 2: Summary of Hypothesis Tests

Hypothesis	Prediction	β	p	Conclusion
H1 (Cross-level)	Larger effect in higher-DLI countries	-0.077	0.020*	Supported
H2 (SES)	Higher-SES benefits more	+0.007	0.147	Not supported
H3 (Trust)	Trust moderates effect	-0.015	0.469	Not supported

Notes: DLI = Digital Literacy Index.

* $p < 0.05$.

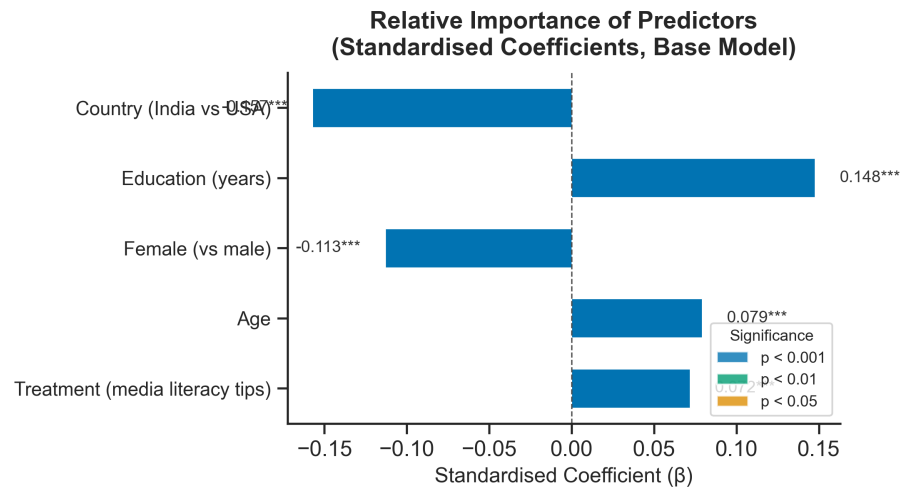


Figure 3 — Standardised beta coefficients: predictor importance

Figure 5: Standardised beta coefficients for predictors of headline discernment, sorted by magnitude and color-coded by significance level. Education and country (India vs. USA) are the strongest predictors.

6 Conclusion

This study provides experimental evidence that the effectiveness of a digital media literacy intervention is stratified by national context. The same intervention that improves headline discernment by 0.14 standard deviations in the United States produces only half that effect in India. This finding carries a sobering implication for global misinformation policy: the countries that most need effective interventions — those with lower levels of digital literacy, weaker educational infrastructure, and less developed media systems — may be precisely those where standard, unadapted interventions are least effective. From a sociological perspective, this pattern is consistent with the knowledge gap hypothesis (?), which predicts that information campaigns disproportionately benefit those with higher existing capacity to process information, thereby widening rather than narrowing inequalities. At the individual level, we find no evidence that the intervention exacerbates SES-based gaps within countries, suggesting that the primary axis of inequality in intervention effectiveness is cross-national rather than within-country.

Several important limitations qualify these conclusions. With only two countries, we cannot disentangle the independent contributions of digital infrastructure, educational systems, media environments, and cultural factors that jointly differentiate the United States and India. The binary treatment condition — tips versus control — limits our ability to test mechanism-specific hypotheses about trust-enhancing versus skepticism-enhancing content. The cross-sectional design cannot speak to the durability of intervention effects over time. Finally, the reliance on online and convenience samples excludes offline and less-connected populations, likely producing conservative estimates of the cross-national gap. Future research should implement multi-country experimental designs with broader country coverage — ideally 15 to 20 countries spanning the full digital development spectrum — and should develop culturally adapted intervention materials with demonstrated measurement equivalence. Such designs are essential for determining whether media literacy interventions can be a tool for reducing global information inequality or whether they will, without deliberate contextual adaptation, reinforce it.

References

- Aalberg, Toril and James Curran. 2013. *How Media Inform Democracy: A Comparative Approach*. Routledge.
- Adjin-Tettey, Theodora Dame. 2022. “Combating Fake News, Disinformation, and Misinformation: Experimental Evidence for Media Literacy Education.” *Cogent Arts & Humanities* 9(1):2037229.
- Ahmed, Saifuddin, Adeline Wei Ting Bee, Sheryl Wei Ting Ng, and Muhammad Masood. 2024. “Social Media News Use Amplifies the Illusory Truth Effects of Viral Deepfakes: A Cross-National Study of Eight Countries.” *Journal of Broadcasting & Electronic Media* 68(5):689–710.

- Altay, Sacha, Rasmus Kleis Nielsen, and Richard Fletcher. 2023. "News Can Help! The Impact of News Media and Digital Platforms on Awareness of and Belief in Misinformation." *The International Journal of Press/Politics* 29(3):578–600.
- Altay, Sacha, Andrea De Angelis, and Emma Hoes. 2024. "Media Literacy Tips Promoting Reliable News Improve Discernment and Enhance Trust in Traditional Media." *Communications Psychology* 2(1):73.
- Apuke, Oberiri Destiny, Bahiyah Omar, and Elif Asude Tunca. 2022. "Literacy Concepts as an Intervention Strategy for Improving Fake News Knowledge, Detection Skills, and Curtailing the Tendency to Share Fake News in Nigeria." *Child & Youth Services* 43(4):325–47.
- Arechar, Antonio A., Jennifer Allen, Adam J. Berinsky, Rocky Cole, Ziv Epstein, Kiran Garimella, Andrew Gully, Jackson G. Lu, Robert M. Ross, Michael N. Stagnaro, Yunhao Zhang, Gordon Pennycook, and David G. Rand. 2023. "Understanding and Combatting Misinformation Across 16 Countries on Six Continents." *Nature Human Behaviour* 7(9):1502–12.
- Bagó, Bence, David G. Rand, and Gordon Pennycook. 2019. "Fake News, Fast and Slow: Deliberation Reduces Belief in False (but Not True) News Headlines." *Journal of Experimental Psychology: General* 149(8):1608–13.
- Chan, Michael, Francis L. F. Lee, and Hsuan-Ting Chen. 2022. "Avoid or Authenticate? A Multilevel Cross-Country Analysis of the Roles of Fake News Concern and News Fatigue on News Avoidance and Authentication." *Digital Journalism* 11(8):1478–98.
- Clayton, Katherine, Spencer Blair, Jonathan A. Busam, Samuel Forstner, John Glance, Guy Green, Anna Kawata, Akhila Kovvuri, Jonathan Martin, Evan Morgan, Morgan Sandhu, Rachel Sang, Rachel Scholz-Bright, Austin T. Welch, Andrew G. Wolff, Amanda Zhou, and Brendan Nyhan. 2020. "Real Solutions for Fake News? Measuring the Effectiveness of General Warnings and Fact-Check Tags in Reducing Belief in False Stories on Social Media." *Political Behavior* 42(4):1073–95.
- Cohen, Meghan Sobel, Karen McIntyre, and Brian Semujju. 2022. "Demographic Differences in Digital News Literacy in East Africa." *African Journalism Studies* 43(4):259–77.
- Faragó, Laura, Péter Krekó, and Gábor Orosz. 2023. "Hungarian, Lazy, and Biased: The Role of Analytic Thinking and Partisanship in Fake News Discernment on a Hungarian Representative Sample." *Scientific Reports* 13(1):178.
- Guess, Andrew M., Michael Lerner, Benjamin Lyons, Jacob Montgomery, Brendan Nyhan, Jason Reifler, and Nee-lanjan Sircar. 2020. "A Digital Media Literacy Intervention Increases Discernment Between Mainstream and False News in the United States and India." *Proceedings of the National Academy of Sciences* 117(27):15536–45.
- Huang, Guanxiong, Wufan Jia, and Wenting Yu. 2024. "Media Literacy Interventions Improve Resilience to Misinformation: A Meta-Analytic Investigation of Overall Effect and Moderating Factors." *Communication Research* 51(6):716–41.
- Lu, Chang, Bo Hu, Meng-Meng Bao, Chi Wang, Chao Bi, and Xing-Da Ju. 2024. "Can Media Literacy Intervention Improve Fake News Credibility Assessment? A Meta-Analysis." *Cyberpsychology, Behavior, and Social Networking* 27(5):307–19.
- Moore, Ryan C. and Jeffrey T. Hancock. 2022. "A Digital Media Literacy Intervention for Older Adults Improves Resilience to Fake News." *Scientific Reports* 12(1):5012.
- Pennycook, Gordon and David G. Rand. 2019. "Lazy, Not Biased: Susceptibility to Partisan Fake News Is Better Explained by Lack of Reasoning Than by Motivated Reasoning." *Cognition* 188:39–50.
- Riad, Abanoub, Barathi Bahavan, and Michal Koščík. 2025. "From Institutional Trust to Digital Literacy: Socioeconomic and Political Determinants of COVID-19 Vaccine Hesitancy Among Czech Adults." *Human Vaccines & Immunotherapeutics* 21(1):2533639.
- Shi, Xiaohan, Jing He, and Gengfeng Niu. 2024. "The Association between Family Socioeconomic Status and Children's Digital Literacy: The Explanatory Role of Parental Mediation." *Adolescents* 4(3):411–24.
- Shirish, Anuragini, Shirish C. Srivastava, and Shalini Chandra. 2021. "Impact of Mobile Connectivity and Freedom on Fake News Propensity During the COVID-19 Pandemic: A Cross-Country Empirical Examination." *European Journal of Information Systems* 30(5):568–87.
- Štětka, Václav, Francisco Brandao, Sabina Mihelj, Fanni Tóth, Daniel Hallin, Danilo Rothberg, Paolo Ferracioli, and Beata Klimkiewicz. 2024. "Have People 'Had Enough of Experts'? The Impact of Populism and Pandemic Misinformation on Institutional Trust in Comparative Perspective." *Information, Communication & Society* 27(14):2603–22.
- Thomas, Pamela Bilo, Clark Hogan-Taylor, Michael Yankoski, and Tim Weninger. 2021. "Pilot Study Suggests Online Media Literacy Programming Reduces Belief in False News in Indonesia." *First Monday* 27(1).

- Tichenor, Phillip J., George A. Donohue, and Clarice N. Olien. 1970. "Mass Media Flow and Differential Growth in Knowledge." *Public Opinion Quarterly* 34(2):159–70.
- Törnberg, Petter and Juliana Chueri. 2025. "When Do Parties Lie? Misinformation and Radical-Right Populism Across 26 Countries." *The International Journal of Press/Politics* 30(2):379–400.
- Tsetsi, Eric and Stephen A. Rains. 2017. "Smartphone Internet Access and Use: Extending the Digital Divide and Usage Gap." *Mobile Media & Communication* 5(3):239–55.
- Wang, Shiwei, G. M. Alam, Karima Bashir, and Miao Lei. 2025. "Reproducing Inequality? Social Capital, Digital Literacy, and SES in Access to Higher Education." *Humanities and Social Sciences Communications* 12(1):626.